Teaching and Mentoring Philosophy

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Teaching Philosophy

My teaching philosophy centers on **connecting first-principles theory to real engineering systems**. The most durable learning, in my experience, comes when students can trace a concept from a governing equation, through a simulation, to a measurement on real hardware. My research provides authentic material for this: dynamometer efficiency maps, thermal measurements, and PWM-loss distributions from industrial prototypes, alongside an in-house Python finite-element code and PyTorch surrogate models that show how analysis, computation, and experiment inform one another. I favor a mixed assessment approach, combining analytical problem sets, simulation-based coursework, and hands-on laboratory work, so that students develop both conceptual depth and practical competence. I also believe physical intuition and computational skill should be taught together, which fits the Physical AI mission of grounding modern AI in physical reality.

Teaching Experience

My formal teaching experience began as a **teaching assistant** on the SEU postgraduate course *Integrated Practice of Electrical Machine Systems*, where I prepared and delivered tutorial material, ran consultation hours, reviewed student project proposals with structured written feedback, and provided mark recommendations to the course coordinator. In parallel, I have provided sustained **technical mentoring** to junior PhD and MS colleagues on multi-physics modeling, finite-element analysis, thermal-network construction, and deep-learning surrogates, reflected in the co-authorship structure of my publications. I recognize that my formal teaching portfolio is at an early stage, and I would value the guidance of experienced colleagues as I develop it.

Courses I Would Contribute

I am prepared to contribute to core computer-engineering teaching and to specialized graduate courses. My hands-on work writing **C/C++ embedded firmware for TI C2000 and STM32 motor-drive controllers**, including real-time control loops, interrupt-driven sampling, and PWM peripherals, equips me to teach **embedded systems** with laboratory content drawn from real control hardware. My background in machine and drive-system modeling, together with that firmware experience, also supports teaching aspects of **computer architecture**, particularly the interface between processors, memory, and real-time peripherals in embedded platforms. At the graduate level I would offer specialized courses in **physics-informed machine learning**, **multi-physics modeling and scientific computing**, and **electric drives and power electronics**, each built around the research-derived tools above. I would be glad to supervise undergraduate capstone and MS projects that connect theory, computation, and hardware.

Curriculum Development

My first priority would be to contribute effectively to the department's existing courses. I would welcome guidance from experienced colleagues on how best to integrate my expertise into established modules, and I would approach curriculum work collaboratively, learning the department's accreditation framework and pedagogical culture before proposing changes. Where the department identifies a need, I could help develop content on **physics-informed modeling of physical systems** and offer practical laboratory resources, for example a **digital-twin exercise** using the Python finite-element code and a deep-learning surrogate, or an **embedded motor-control practical** on the C2000 and STM32 platforms, both of which give students a tangible link between models and measurements.

Mentoring Philosophy

In mentoring junior colleagues at SEU and Nottingham I have developed a preference for **encouraging early ownership** of a research problem while providing close support through regular technical discussion. I try to help students build independent judgment: to question their own results, validate models against data, and communicate clearly. I am attentive to the fact that students arrive with different preparation, backgrounds, and confidence, and I adjust the level of structure I provide accordingly, with the aim that every student I work with can do their best work. I would foster a research-group culture defined by **respect, clear expectations, and fair credit**.

Professional Development

I am committed to developing as a teacher. I would engage with Virginia Tech's teaching-development resources and learn from colleagues about inclusive and effective practices suited to a large, diverse student body. I see teaching and research as mutually reinforcing: the questions students raise sharpen my research, and the tools my research produces enrich the classroom.